SUPPLY AND DEMAND



After studying this chapter, the students should:

- Explain how decisions are made on what to produce, how to produce and for whom to produce, how prices act to allocate resources within an economy
- Explain Consumer behaviour and demand
- Draw standard demand and supply curves
- Explain Price determination
- Explain why prices change from time to time, the main influences of demand and supply
- Distinguish between a change in demand or supply, and a change in the quantity demanded and supplied
- Explain why and how the government intervenes
- Explain the effects of government intervention

1.0 INTRODUCTION

This chapter deals with how the free market Economic systems deals with the allocation of scarce resources, making choices on what, how and for whom to produce. This emphasis is on the market for goods and services. However, the factor market, which is the market for factors of production, land, labour, capital and enterprise, with the corresponding rewards, rent, wages, interest and profit respectively, works in almost a similar way.

A market is where buyers and sellers meet, it does not necessarily mean a geographical location. What determines what and how much of anything to produce is the **price,** and price results from the operation of **demand** by buyers and **supply** from sellers.

In a free market, prices, which are basically determined by demand and supply, combine to solve the problem of resource allocation. Prices act as a signal of what people want to buy, indicating to producers where their scarce factors will most profitably be utilized.

1.1 DEMAND

Individual demand must be differentiated from wants or desires. Demand refers to the willingness by consumers to own goods, and it must be backed by money, it is therefore, qualified as **effective** demand. This is the quantity of a product or service that consumers are willing and able to buy at a given price. Emphasis is not only willingness, but this must be supported by the ability to pay. **Market demand** is the **total** quantity, which all customers are willing and able to buy at a particular price.

1.2 THE DEMAND SCHEDULE

There is an inverse relationship between the quantity demanded and price, the amounts that a consumer is willing and able to purchase at various prices at any given time tends to be high at low prices, and low at high prices.

Below is Mr Banda's demand schedule for mangoes in the month of November.

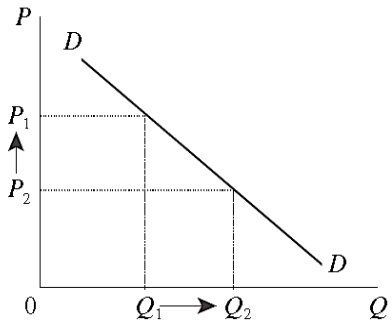
Price (K)	Quantity demanded (units)
1 000	0
800	3

500	4
300	6
200	10

1.3 DEMAND CURVE

When the data above is plotted into a line graph, a demand curve is produced.

FIG 3: DEMAND CURVE



The above figure shows that there is an inverse relationship between quantity demanded and the price of the commodity, for a normal good. But in a situation where more of a good is demanded when its price increases then such a good is said to be a **giffen good**.

A ***giffen good*** therefore is that good whose demand increases as its price increases where as this good is not a luxury. The reason for this positive relationship between the price and demand of the commodity is that; as the price increases, the consumer is left with no other alternative but buying this good in order to satisfy his basic needs.

A 'normal' demand curve slopes downwards from left to right, due to changes in price. A change in price never shifts the demand curve for any good, it results in a movement along a demand curve. This is **a change in the quantity demanded.**

The Law of Demand

- The ***law of demand*** states that the quantity demanded of a good or service is inversely related to the selling price, *ceteris paribus (*all other determinants remaining unchanged)
- The law of demand is a misnomer. This is because laws are assertion of facts. They predict events with certainty
- However, the 'law of demand' does not always predict with certainty. It is more of a theory than a law
- Theories are predictions without complete certainty
- Symbolically, the law of demand can be summarized as
    - $QD = f(P)$ and
    - $\frac{dQ_D}{dp} < 0$
- The first formula states that the quantity demanded is functionally related to price
- The inequality states that quantity demanded and price are inversely related
- This is illustrated by the demand curve

Determinants of Market Demand

i) **Price of the commodity**: this is considered to be the major factor of quantity demanded of any good or service. Indeed the relationship between the price of the good in question (under investigation) and the corresponding quantity demanded gives rise to a demand curve. Generally, a consumer is likely to demand more of a commodity when its price falls and

less of the commodity when its price rises. This is true for a **normal good**. This is because given fixed income of a consumer of a consumer; she can be able to purchase more units of a commodity only when its price falls *ceteris paribus*.

ii) **Income of the consumer**: Another major factor of demand is the consumer's income. Time and gain, we observe richer individuals purchasing goods such as fuel, meat, telephone, credit card, and drinks to mention but a few in large quantities while the poor usually buy in smaller quantities of any good. And when one's level change, his/her expenditure pattern change as well. Usually, more of a normal commodity will be demanded when the consumer's income increases and less of it will be demanded at lower incomes. However for an inferior commodity, less of a commodity is demanded at higher income levels of a consumer and more of commodity will be demanded when the consumers' income falls.

iii) **Price of related commodities**: Related commodities can either be **complements or substitutes**. Complimentary commodities refer to those commodities that have to be consumed at the same time e.g. Coffee and sugar, car and fuel, beef and tomatoes, beef and salt, etc. On the other hand **substitute commodities** are those that may not be consumed at the same time or those where consumption of a commodity may stop the consumer from consuming the other. For instance riding a motorcycle or bicycle, taking tea or coffee, eating fish or beef. The last combination (i.e. fish and beef) suggests that substitute commodities are not always necessary mutually exclusive.

iv) **Tastes and preferences of the consumer**: Many commodities are consumed due to consumer's tastes and preferences and any change in tastes may adversely affect the quantity demanded. Such goods may include food, fashions, cars, drinks and may others. Preferences can either be natural or under influence from peers. Generally, however, a consumer's tastes and preferences can change at any one time and will affect the quantity demanded of the good in question. For instance supposing that tastes change in favour of a commodity, then, more of that commodity will be demanded when the tastes and preferences change in disfavour of a particular commodity.

v) **The expected price of the commodity**: If the future price of the commodity in question is expected to fall, consumers will demand less of the commodity now. On the other hand, more of the commodity will be demanded or purchased when its price is expected to rise. This is however, common with goods that can easily be stored for future consumption, price expectation matters a lot on the decision to consume now.

vi) **Season of the year**: Some goods tend to be consumed during certain seasons, so their demand is likely to rise during such seasons. For instance, during wet seasons people demand more umbrellas and rain coats. Also during Christmas or Edi season, more Christmas cards or Edi cards are demanded.

vii) **Religions and culture**: the quantity demanded of a commodity, may also be affected by religion and culture of the consumers. For instance, the demand for pork is likely to be low in a society where there are many Moslems. Also in some African societies, certain commodities are prohibited, meaning that the consumption is a taboo; while some cultures promote the consumption of certain commodities or certain fashions of goods (such as dresses such as *Citenge* in the central region of Africa).

viii) **Sex**: this is also another factor of demand. It is worth noting that certain commodities are consumed by a particular sex group. For instance shirts, skirts, bras and others. Particularly,

the demand for bras is likely to be high in a society where there are more female than male, keeping other factors constant. So we would expect the demand of bras to be high in a girls' hostel and that of shirts to be high in a boys' hostel.

ix) **Level of education**: There are certain commodities that are consumed by a certain group of people, where the grouping is based on education levels. For instance the demand for scholastic materials such as pens, reams of paper, exercise books etc is likely to be low in a place where very few people go to school and vice versa. Therefore, it would be unprofitable if one located a stationary shop in a place where there are no schools.

x) **Marital status** : There are certain goods that are consumed mainly by married couples for instance wedding rings, some style of dress for instance *Citenge** .

xi) So the demand of such commodities is likely to be high in a place where there are many married people. Indeed it would be unprofitable if one located a *Citenge* shop in a secondary school premises

xii) **Age**: this is also an important factor of demand. There are certain commodities that are consumed by certain age brackets. Such goods Include; hairstyles, dresses, alcohol and others. We find that the demand for a certain hair style, jeans, "mini-skirts" is high among the teenagers than among the old "*ceteris paribus*".

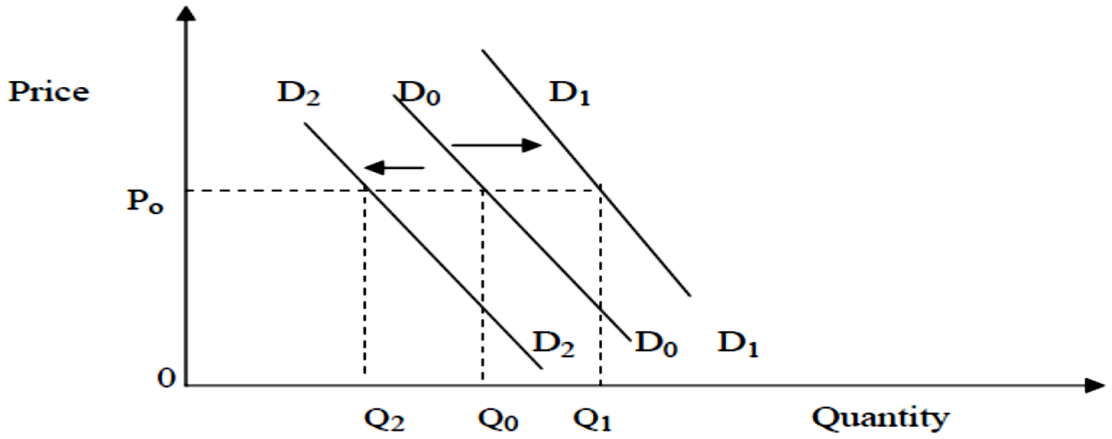
xiii) **Number of consumers**: Quantity demanded can also be influenced by the size of the population, but more so by the income distribution of the society. Producers will always want to locate in places that are more populated than in sparsely populated places. This is more major reason why countries integrated to form trade blocks. Large markets tend to attract foreign investors because they expect higher turnovers. Likewise on the local scene, many traders tend to locate their businesses in towns other than rural areas because of high population hence higher demand in towns than rural areas.

xiv) **Location of consumer**: the location factors may either be geographical or due to level of economic activity. For instance if individuals are located in hot aces such as places at lower altitudes tropical places e.g. Accra, Dar es Salaam, Mombasa, Etc goods such as blankets may face a lower demand while Air conditions face a higher demand. On the other hand, we would expect the demand for sweaters coasts to be high in the cooler parts of a country than in hotter areas. Similarly, the demand for electrical appliances will be very low in areas where there is no electricity and it will be high in a location where there is electricity. As regards to the level of Economic activities, we would for instance expect the demand for fertilizers and pesticides to be high in crop growing areas than in other areas.

Change in Demand

This occurs when there is a change in any other factor that determines quantity demanded other than the price of the commodity in question. Such factors may include change in the income of the consumer, price of a related commodity, tastes of the consumer and others factors. Change in demand is shown by a shift in the demanded curve upwards to the right or downwards to the left, depending on the direction of change of the concerned variable

Change in Demand



At a constant price Po a change in other factors which affect demand of the commodity, shifts the demand curve to either D_1 or D_2 from D_0 depending on the direction of change in the variable. If the change increases demand then the demand curve shifts outwards to D_1 but if the change reduces demand of the commodity, then the demand curve shift inwards to D_2. For instance increase in the consumers' income other factor kept fixed would shift the demand curve outwards to the right and vice versa. Also if the consumer's tastes change in favour of the commodity, its demand curve shifts outwards to the right and vice versa.

Supply Analysis

Introduction

By the term quantity supplied, we refer to the quantity of a commodity which a producer is willing to bring to the market at the prevailing price. The factors that influence supply differ from one commodity to another. For instance those that determine the supply of beef are different from those of maize and also different from those of an industrial commodity such as price of the commodity, cost of production and others.

Quantity supplied

Quantity supplied, is the quantity of a commodity produced and available for sale in a market at any given time and price. Quantity supplied is determined by several factors, which include the following:

1. Price of the commodity
   i) Cost of producing the commodity
   ii) Price expectations
   iii) Price of other commodities (substitutes or complements)
   iv) Taxes on the commodity
   v) Number of producers
   vi) Level of technology
   vii) Natural hazards (in the case of agriculture products)
   viii) Government policy
   ix) Political climate and
   x) Availability of factors of production.

Price of the commodity
This is the major determinant of quantity supplied of any commodity. Producers are willing to put more of a commodity on the market when its goes up and less when its price falls. This is also known as the law of supply. Economic researcher has shown that supply is more responsive to price than to any other factor. In other words, to induce producers to produce more, the price for the commodity should be raised.

Cost of production
During a production process, several factor inputs are employed, these include; raw material, labour, energy, capital and others. For a producer to use factors there must be payment (reward for the factor of production) for them. These rewards constitute costs.
The extent of costs should be in line with the consumer's ability to pay for the production commodity and also in line with the producers' ability to afford cost. Nevertheless, cost is a disincentive to production. Accordingly, the higher the cost of production, the less the quantity supplied of a commodity at any period and vice versa.

Price Expectation
A producer, who involves himself in market research, is always out to know the trend of prices for his commodity. The expected price is affected by the demand which in turn is influenced by the number of consumers, which may arise from advertising, or it may change due to periods of the year for instance Christmas season, students going back to schools, The producer will supply more now, if he expects to fall in future and vice versa.

Price of other goods
Quantity supplied of commodity may also be influenced by the price of related goods; such goods may either be complements or substitutes. If the price of substitute goods is low, the producer will supply less but if the price of the complementary goods is low, she will supply more.

Taxes on the commodity
Increase in taxes raise the cost of production, thus negatively impact on production. Therefore, at high levels of tax, a producer tends to supply less of a commodity to the market.

Number of producers
Indeed in a market where there is single producer also called monopoly the level of output is less, while where there are many producers, i.e. perfect competition, the level of output is high. Therefore, the smaller the number of producers the lower the level of output, and the greater the number of producers the greater the level of output.

Level of technology
The level of technology influences the degree of misuse (waste) of raw materials, and production capacity of the firm. Indeed remote technology yield less output hence less is supplied while advanced technology leads to little inputs wasted, greater production capacities hence more output supplied to the market.
Natural hazards
These are uncontrollable factors that may include; floods, drought and a general change in climate. Such factors affect production especially of agricultural commodities although there may be less

or no effect on industrial commodities unless the key factor inputs are agricultural produces. Accordingly, where ever such calamities occur, less output shall be supplied.

Government policy
Government can encourage or discourage production of particular commodity. Government can openly campaign for more production of a particular good by either subsiding its production, providing seeds in case of crops, and ensuring availability of ready market either local or international. In such a case more output shall be supplied. On the other hand, government can out-right ban production of a particular good for instance cocaine and impose penalties to those who promote the production of such a good.

Political climate
Political climate affects all sorts of production, weather agricultural or industrial. In areas where there are wars, little or no production takes place. Accordingly, there is less supply of any commodity in such a region. Similarly conducive political environment encourages production hence more output can be supplied.

Availability of factors of production
Availability of factor inputs directly affects the level of production particularly for industrial commodities where factor of inputs are less available, less output is produced industrial commodities where factor of inputs are less available, less output is produced hence supplied.

Season; (time period of the year)
Some commodities are supplied in particular season. For instance Christmas cards, success cards or seasonal cards in general. Also many agricultural crops tend to be harvested around the same period. Such as beans, coffee, maize, millet, coffee etc. therefore quantity supplied of such commodities is high in particular season and low the rest of the periods.

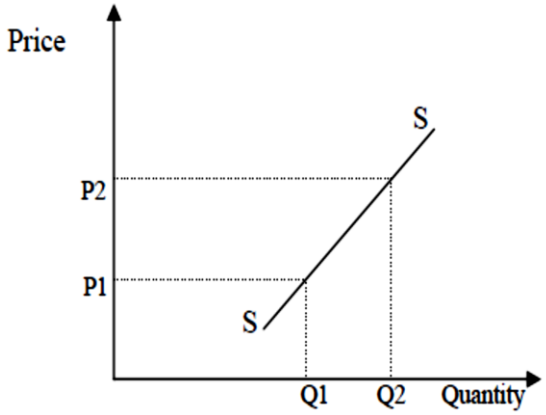
The supply curve

From the above discussion, we have seen that there are many factors that influences the quantity supplied of any commodity. However, if we now assume that other factors do not affect the level of supply other than price of the commodity so that we investigate the affect of the changes in the price of the commodity on the quantity supplied. If this relationship is expressed in the table from, we call such a table the **supply schedule**. In other words, a supply schedule is a table that shows the relationship between price of a commodity and quantity supplied at the various prices.

Table 2.3 The supply schedule

Below is Ms Chanda's supply schedule in the month of November.

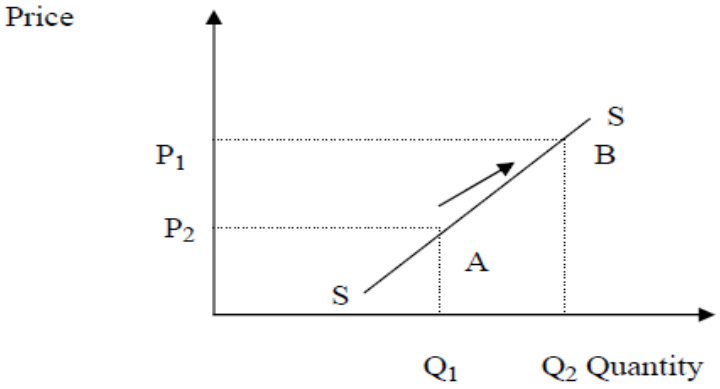
Price (K)	Quantity supplied (units)
1 000	0
800	3
500	4
300	6
200	10



The supply schedule shows the producer's willingness to supply different quantities of the commodity at the given prices to the market. Generally, a producer is much willing to supply more when price is higher. When the information from the supply schedule is transferred to a graph, we get a **supply curve**. A supply curve therefore is a graphical representation of a supply schedule.

Generally, when price is low, the producer is willing to supply less to the market and at higher price the producer supplies more. This leads to the law of supply that states that "the higher the price of a commodity, the greater the quantity supplied ceteris paribus". Accordingly, a normal supply curve ha a positive slope (i.e. is up-ward sloping). However, there are instances when the supply curve is down-ward sloping thus contradicting the law of supply. Such a down-ward sloped supply curve is usually referred to as an abnormal supply curve.

Change in quantity supplied

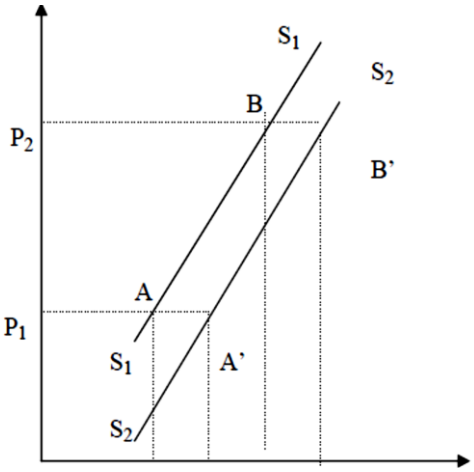


From the proceeding section, the law of supply is based on the assumptions that other factors of supply do not change, such that when price of the commodity changes, quantity supplied also change in the same direction as the price change. This is referred to as **change in quality supplied** and is shown by a movement along the same supply curve. In other words, change in quantity supplied is caused by a change in the price of the commodity other factors kept constant.

An increase in price from P_1 to P_2 leads to an increase in quantity supplied from Q_1 to Q_2. This is equivalent to a movement from point A to B along the same supply curve.

Change in supply

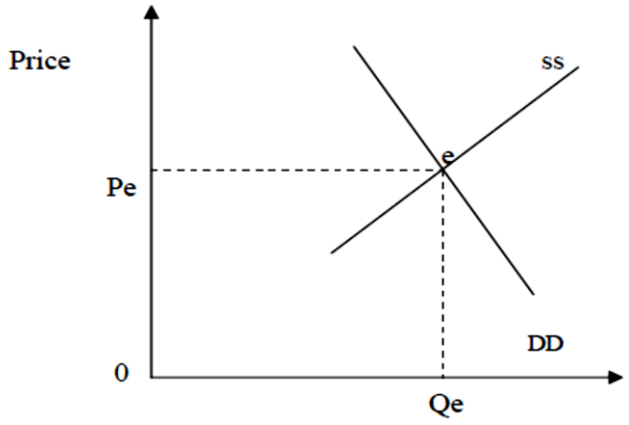
If we now allow other factors of supply to change while the price of the commodity is kept fixed, then more quantity shall be supplied at the same price if there is an increase in the factor that favour supply, such as increased number of producers, better technology, availability of raw material etc. On the other hand, less shall be supplied at a given price when there is an increase the factor that



disfavour production such as; cost of production, natural hazards, fall in the number of producers etc. This is known as **change in supply** and is shown by a shift in the supply curve

Equilibrium in a market

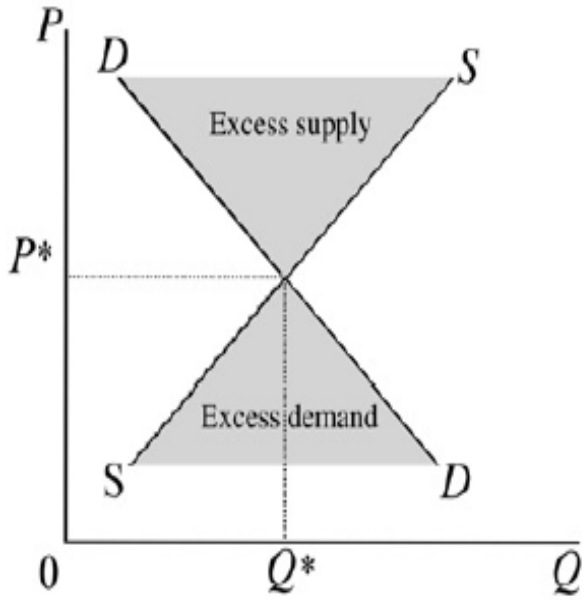
A market is said to be in equilibrium when quantity supplied is equal to quantity demanded. It occurs when both the consumer (demand curve) and producer (supply curve) co-exist interact using forces of demand and supply (market forces) to yield the market price and quantity (equilibrium price and quantity). At that point, there is no tendency of any of the variables to change since equilibrium is considered to a state of stability.



Graphically the point of equilibrium is defined by the point of intersection of the market demand curve and supply curve. The price and quantity at the equilibrium point are referred to as equilibrium price and equilibrium quantity respectively.

In the above figure, Pe is equilibrium price and Qe is equilibrium Quantity. In this case the forces of demand and supply are said to be in equilibrium. This simply means that both the producer and consumer are equally satisfied because whatever the consumer demands is supplied by the producer and whatever is supplied is demanded hence there is no **rationing** in the market.

Interaction of Demand and Supply



i) Concepts of demand and supply can be used to explain the functioning of the market mechanism

ii) At price P*, the quantity of a good or service that buyers are able and willing to buy is precisely equal to *Q*, the* amount that firms are willing to supply

iii) At a price below *P*, the quantity* demanded exceeds the quantity supplied

iv) Consumers will bid among themselves for the available supply of *Q, which will drive up the* selling price

v) Buyers who are unable or unwilling to pay the higher price will drop out of the bidding process

vi) At the higher price, profit-maximizing producers will increase the quantity supplied

vii) Excess supply of products will push the price down until the equilibrium is reached

Example: The market demand and supply equations for a product are:

a. *QD = 25 - 3P*
b. *QS = 10 + 2P*
c. where *Q is quantity and P is price.*

What are the equilibrium price and quantity for this product?

d. *25 - 3P = 10 + 2P, P = 3*
e. *Q* = 25 - 3(3) = 10 + 2(3) = 16*

The Rationing Function of Price

- The rationing function of prices refers to the increase or a decrease in the market price to eliminate a surplus or a shortage of a good or service
- The rationing function is considered to be a short-run phenomenon because other demand determinants are assumed to be constant
- While price rationing is a fundamental characteristic of market economies, it is not the only way to allocate goods and services that are in short supply
- Alternative rationing mechanisms are necessary when the market is constrained from performing this function
- Under what circumstances might the market price fail to increase to eliminate a shortage

1. Price Ceilings

i. A price ceiling is a maximum price for a good or service that has been legally imposed on firms in an industry
ii. The rationale for price ceiling if fairness
iii. Price ceiling is justified sometimes, but more often than not result in unintended negative consequences

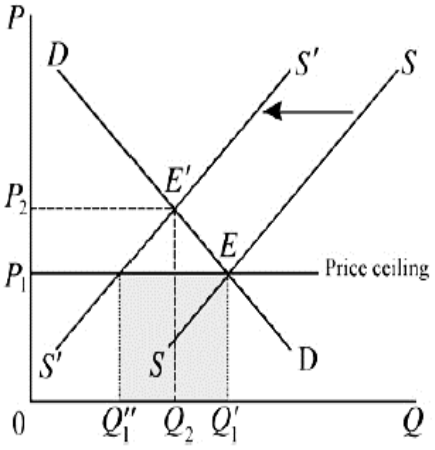


Figure on the left depicts excess demand, Q" – Q'
As Zambia shifted resources to producing military goods and services, fewer commodities were left for domestic consumption.
Without price controls, the equilibrium quantity would have fallen from *Q1 to Q2 and price from P1 to P2.*
The rationing function of prices would have guaranteed that only the well-to-do had access to available, non-military commodities.
In the interest of "fairness," the government imposed ***price ceilings, P1***
The imposition of a price ceiling means shortages will not be automatically eliminated by increases in price

- Some other mechanism for allocating available supplies of consumer goods like ration coupons were imposed
- Coupons entitle the holder to purchase a given amount of a particular good or service during a given time period

Example: The market demand and supply equations for a product are:

$Qd = 300 - 3P$

$Qs = 100 + 5P$

where *Q is quantity and P is price*

What are the equilibrium price and quantity for this product?

$300 - 3p = 100 + 5P, P* = K25$

$\boldsymbol{Q* = 300 - 3(25) = 100 + 5(25) = 225}$

- Suppose that an increase in consumer income resulted in the new demand equation

$$Qd = 420 - 3P$$

What are the new equilibrium price and quantity for this product?

$$420 - 3P = 100 + 5P, P* = 40$$

$$\boldsymbol{Q* = 420 - 3(40) = 100 + 5(40) = 300}$$

- Suppose the government enacts legislation that imposes a price ceiling equivalent to the original equilibrium price K25. What is the result of this legislation?

At the price ceiling of *P = K25 the quantity demand is*

$$Qd = 420 - 3P = 420 - 3(25) = 345$$

At the price ceiling the quantity supplied is

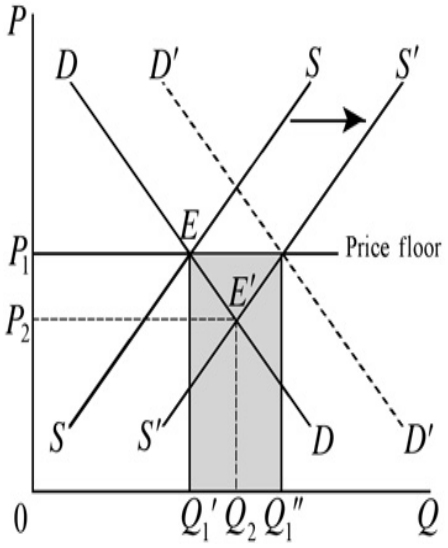
$$Qs = 100 + 5P = 100 + 5(25) = 225$$

- Based on these results, there is a shortage in this market of

$$\boldsymbol{Qd - Qs = 345 - 225 = 120}$$

Price Floors

A price floor is a legally imposed minimum price that may be charged for a good or service



Price floors are designed to keep prices from falling below some legal minimum
Price floors are the opposite of price ceiling
It is common in the agricultural prices and minimum wages
The figure shows an excess surplus of a commodity due to an increase in supply
In the absence of a price floor, the equilibrium price of the product would have fallen from *P1 to P2 and the equilibrium* Quantity increased from *Q1 to Q2*
In the labour market, price floors in the form of minimum wage legislation are designed to provide unskilled workers with a "living wage,"
But the result is usually an increase in the unemployment rate of unskilled labour

- **Price floors normally create problem of surplus**
- Government has two policy options to maintain P1 price
- 1. Pay certain farmers not produce so that supply curve does not shift from *S* to *S'*
- 2. Enter the market and buy the produce (effectively shifting the demand curve from *D* to *D'*
- In either case, the taxpayer pays the bill

Example

Consider the following demand and supply equations for the product of a perfectly competitive industry:

- $Qd = 25 - 3P$
- $Qs = 10 + 2P$

- a. Determine the market equilibrium price and quantity algebraically
  - $Qs = QD;\ 25 - 3P = 10 + 2P, P* = 3$
- The equilibrium quantity is determined by substituting the equilibrium price into either the demand or the supply equation

$$Qd* = 25 - 3(3) = 25 - 9 = 16$$
$$Qs* = 10 + 2(3) = 10 + 6 = 16$$

- Suppose that government regulatory authorities imposed a "price floor" on this product of $P = K4$.
- *What would be the quantity supplied and* quantity demanded of this product?
- At a price floor of $P = K4$*, the quantity demanded and quantity supplied* can be determined by substituting this price into the demand and supply equations and solving:

$$Qd = 25 - 3(4) = 25 - 12 = 13$$
$$Qs = 10 + 2(4) = 10 + 8 = 18$$

- Since ***Qs > Qd****, these equations describe a situation of excess supply* (surplus) of 5 units of output

- Free markets determine not only final distribution of final goods and services (price rationing – short-run phenomena);
- But also what goods and services are produced and how productive resources will be allocated for their production in the long-run (allocating function)
- The allocating function of prices refer to the process by which productive resources are reallocated between and among production processes in response to changes in the prices of goods and services
- ***The interaction of supply and demand is the primary mechanism for the allocations of goods, services, and productive resources in market economies***